

Vaginitis — Diagnosis & Treatment Reference

Vulvar/vaginal itching, burning, abnormal discharge/odor · TeachIM.org (Dec 2022, optimized Jun 2026) · CC BY-NC 4.0 · educational use only — clinician-review before teaching

1 · Pathophysiology & pH

Cause	Mechanism	pH
BV (15–50%)	↑ anaerobes (<i>Gardnerella</i> , <i>M. hominis</i>), ↓ lactobacilli	>4.5
Yeast (75% lifetime)	<i>Candida</i> overgrowth (<i>albicans</i> 80–92%, <i>krusei</i> , <i>glabrata</i>); lactobacilli unchanged	Normal 4.0–4.5
Trichomonas (STI)	Motile <i>T. vaginalis</i> ; ↓ lactobacilli	>4.5 (often >5)
GSM/atrophic	↓ estrogen → ↓ lactobacilli	>5.0–5.5

- Normal flora: 70–90% *Lactobacillus* → lactic acid → pH 3.8–4.5 (estrogen-promoted). **Yeast is the one with normal pH.**

2 · Differentiate (H&P)

	BV	Candida	Trich
RF	New/multiple/same-sex partners; douching; smoking	Antibiotics; DM; luteal; immunosupp	New/multiple partners; smoking; no barrier; STIs
Dx	Thin homogeneous; gray/white	Thick "curd-like," white, no odor	Copious, frothy, green-yellow
Exam	No inflammation; ± fishy odor	Erythema/excoriation; itch	Vulvar erythema/edema; strawberry cervix

3 · Diagnostics

	BV	Candida	Trich
pH	>4.5	Normal	>4.5, often >5
Whiff	+	–	±
Wet prep	Clue cells	Hyphae/budding (KOH)	Motile trichomonads + ↑WBC (Sn 50–60%)
Best test	DNA assay / gram stain (Nugent); Amsel ≥3/4	DNA assay; culture	NAAT/PCR (most Sn & Sp)

- **Amsel (3/4):** thin homogeneous discharge; pH >4.5; + whiff; clue cells ≥20% epithelial cells.
- Symptoms are nonspecific — offer exam + pH + microscopy and/or DNA assay/NAAT to all symptomatic patients; **exclude STI cervicitis (GC/chlamydia)**. Negative wet prep does NOT rule out trich.
- BV + yeast can coexist (↑ pH with both clue cells & hyphae).

4 · Treatment

Cause	First-line	Pregnancy
BV	Metronidazole 500 mg PO BID ×7 d, or 0.75% gel ×5 d, or clinda 2% qHS ×7 d / 300 mg PO BID ×7 d	Topical or oral metronidazole
Yeast	Fluconazole 150 mg PO ×1, or topical azole (miconazole/clotrimazole)	Topical azole ×7 d
Trich	Metronidazole 500 mg PO BID ×7 d (or tinidazole 2 g ×1); treat partner(s) ; no topical	Metronidazole 500 mg BID ×7 d
GSM	Vaginal estrogen; moisturizers/lubricants	—

- **Trichomonas = STI:** treat partner (men 2 g ×1), avoid sex until both treated, re-test at 3 mo.
- **Risk reduction (recurrence):** avoid douching, clean sex devices, barrier protection; glycemic control + avoid unnecessary antibiotics (yeast); smoking cessation (BV/trich).

5 · Newer context (core unchanged)

- Deck matches 2021 CDC STI Guidelines (current); trich women: multi-dose > single 2 g.
- **Secnidazole (Solosec)** 2 g ×1 — single-dose for BV (2017) & trich (2021).
- Recurrent VVC: **ibrexafungerp** (Brexafemme) & **oteseconazole** (Vivjoa, FDA 2022, first RVVC drug).

Sources: CDC MMWR 2021 (RR-70/4); ACOG PB 215; Marnach Mayo Clin Proc 2022; FDA approvals (Solosec, Brexafemme, Vivjoa). Adapted from TeachIM.org under CC BY-NC 4.0.